Lesson 9 Teacher's Key: On the Scoreboard

Scratch - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

That a genuinely "real" UI can be entirely free - Scratch's built-in variable watchers are not a placeholder or a workaround, they're the actual, correct tool for this job in this engine, the same way draw_gui is GameMaker's correct tool.

Level 1 - Full Worked Solution

- score is a second "for all sprites" variable, exactly parallel to lives and enemies_remaining.
- The score change lives inside GameManager's existing "when I receive [enemy hit]" script, alongside the enemies_remaining change - one broadcast, two consequences, both handled on the one object that already owns this game's shared state. The Enemy sprite still only ever broadcasts enemy hit; it never touches score itself.
- Checking the watcher checkboxes is the entire "UI implementation" - there is no additional code, no additional sprite, nothing else to build.

Highlighted lines are new this week; everything else is code you already have.

```
// GameManager (hidden sprite) - existing script:
when I receive [enemy hit]
change enemies_remaining by -1
change score by 10
if enemies_remaining = 0 then
    broadcast [you win]

// Enemy sprite: no changes - it still only broadcasts [enemy hit] and
never touches score.

// Variables palette: tick the checkbox beside score AND beside lives so
both show on stage.
```

Why This Structure

- This is a valuable contrast to draw next to Unity/Unreal's much more involved setup this week: sometimes the simplest tool really is the right tool, and recognizing that (rather than assuming more setup = more correct) is itself a production skill.
- Keeping score and enemies_remaining changes in the same GameManager script mirrors the reuse principle established since Lesson 6 - anything that's a direct, simultaneous consequence of the same broadcast belongs in the same place, on the one dedicated manager sprite.

Level 2 / Level 3 Talking Points

- Ask students who also work in another engine this week to compare setup effort - Scratch's total UI work is two checkboxes, versus several new nodes/objects elsewhere. Worth naming explicitly as a tradeoff of using a block-based, watcher-driven engine.
- Level 3 visual: "large readout" mode is worth demonstrating live - it's a genuinely different-looking display from the default small watcher, useful for students who want their HUD to feel more like a real game.

Level 2 - complete code:

```
// Level 2 is verification only - the Level 1 code is already correct, so
no lines change.
// Checks: 3 kills read exactly "Score: 30"; an escape moves lives but
never score;
// both numbers are correct the instant the game starts, not only after
the first kill or escape.
```

Level 3 - complete code (all goals combined):

Highlighted lines are the Level 3 additions on top of the Level 1 code.

```
// GameManager - when I receive [enemy hit] (Level 3 variety):
change enemies_remaining by -1
change score by (pick random 5 to 15)
if enemies_remaining = 0 then
    broadcast [you win]

// Level 3 visual: right-click the score or lives watcher on stage and
choose "large readout"; drag the watchers to a free corner.
```

Common Student Mistakes

- Forgetting to check the watcher checkbox for one of the two variables (usually lives, since it was added in a prior lesson and may already be hidden by default).
- Adding "change score by 10" to GameManager's wrong script (the "when I receive [enemy escaped]" script instead of "when I receive [enemy hit]").
- Confusing the show/hide checkbox in the Variables palette with the sprite's own show/hide - these are unrelated; the variable watcher's visibility is independent of any sprite's.